Computer Networks Lab 06

Course: Computer Networks (CL3001) | Semester: Fall 2026
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Lab Instructions & Note

- Introduction to Telnet & configuration of Telnet in Cisco Packet Tracer.
- Introduction to SSH & configuration of SSH in Cisco Packet Tracer.

TELNET & SSH

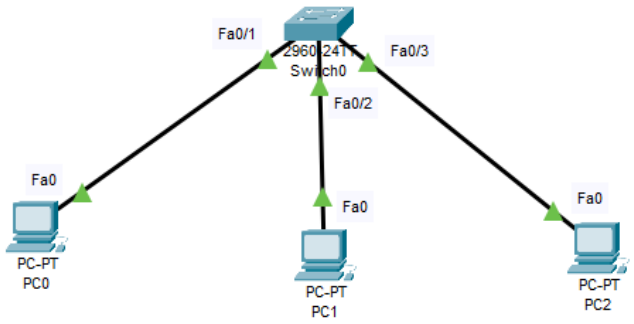
1. Introduction to Telnet

A terminal emulation program for TCP/IP networks such as the Internet. The Telnet program runs on your computer and connects your PC to a server on the network. You can then enter commands through the Telnet program and they will be executed as if you were entering them directly on the server console. This enables you to control the server and communicate with other servers on the network. To start a Telnet session, you must log in to a server by entering a valid username and password. Telnet is a common way to remotely control Web servers.
To telnet means to establish a connection with the Telnet protocol, either with command line client or with a programmatic interface.

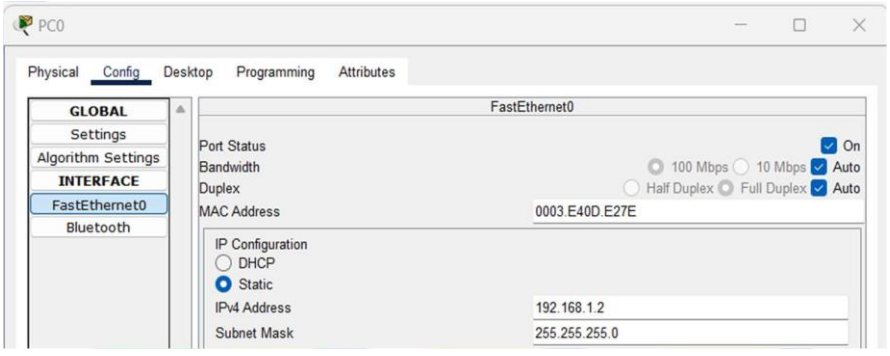
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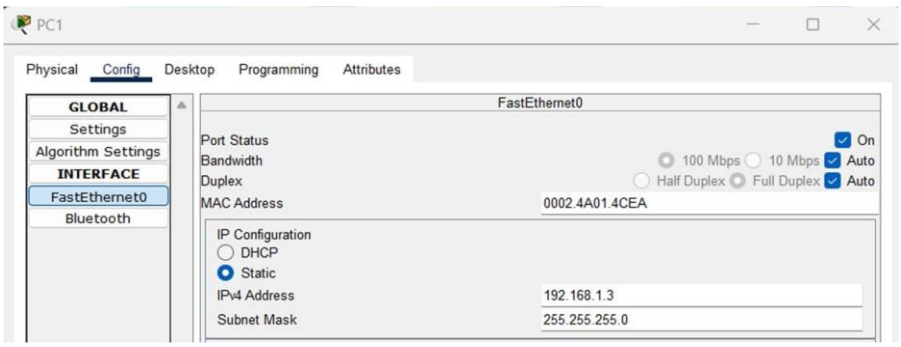
2. Configuration of Telnet

Below are the steps for Telnet Protocol. Follow the procedure for the configuration of Telnet Protocol.
Step 1: Create the following topology



Assigning IP to PCs:







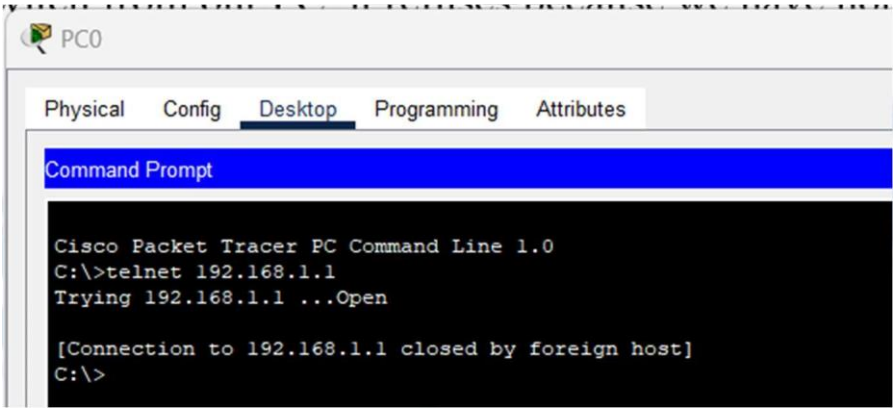
Step 2: As, by default, all PCs are in vlan. We will create a virtual interface on switch with vlan 1 as follows:

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

Step 3: Now, try to telnet the switch from our PC, it refuses because we have not applied authentication on the switch yet.

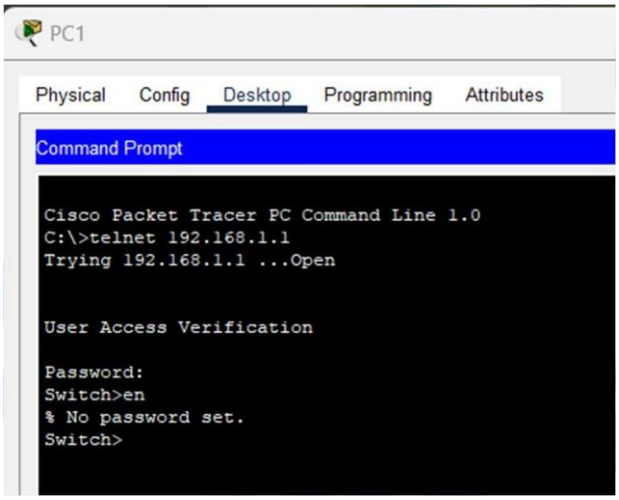


For Secure Shell Server (SSH) and FTP services each Telnet, SSH, or FTP session requires one vty-line. You can add security to your system by configuring the software to validate login requests:

```
Switch(config-if)#exit
Switch(config)#line vty 0 15
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

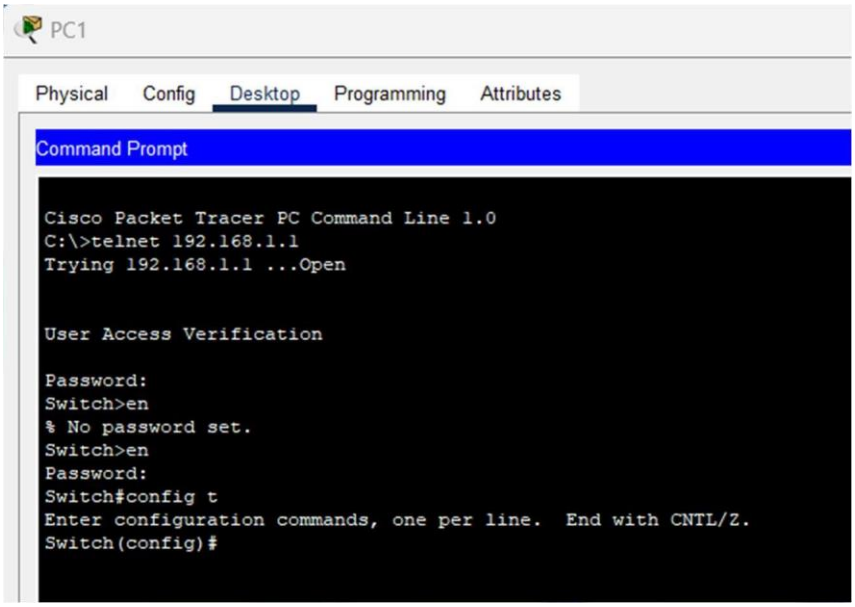
Now, we can easily telnet. But it does not let us go in the switch enabled mode because we have not set the password on the switch yet.



Let's apply password on the switch enabled mode.

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#enable password cs
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Now, we can go inside Switch configuration mode from our pc.



3. Introduction to SSH

Secure Shell or Secure Socket Shell is a network protocol. It is an application layer protocol that is in the 7th later of the Open Systems Interconnection (OSI) network model. It also refers to the suite of utilities that implements the SSH protocol.
Secure Shell also supports both password and key-based authentication. Password-based authentication let users provide username and password to authenticate to the remote server.
A key-based authentication allows users to authenticate through a key-pair. The key pairs are two cryptographically secure keys for authenticating a client to a Secure Shell server.
Furthermore, the Secure Shell protocol also encrypts data communication between two computers. It is extensively used to communicate with a remote computer over the Internet.

4. Configuration of SSH

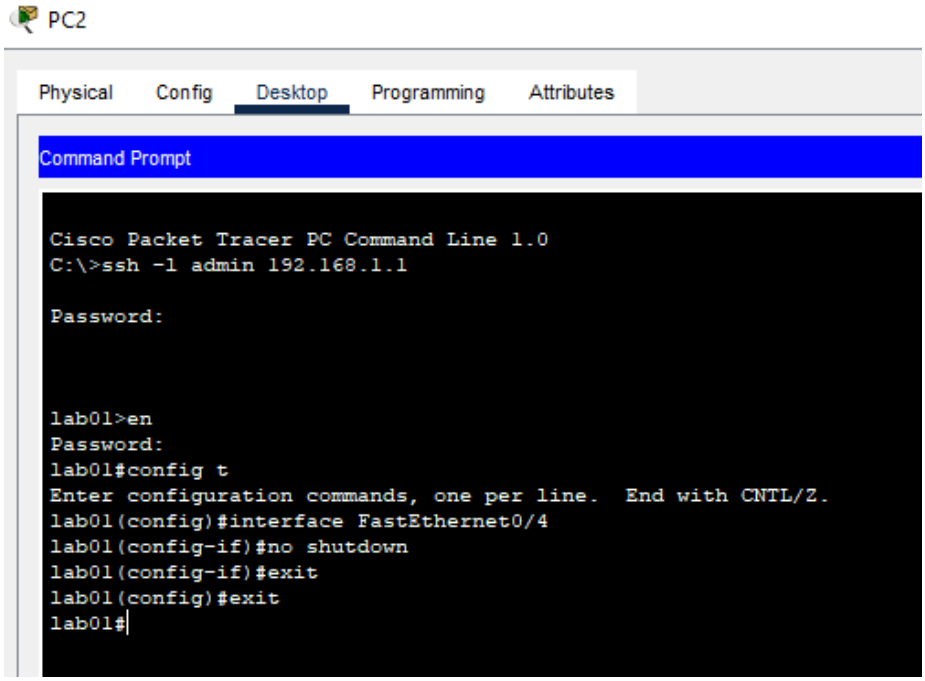
Taking the same topology as mentioned in telnet.
Below are the steps for SSH Protocol. Follow the following procedure for the configuration of SSH Protocol

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname lab01
lab01(config)#ip domain name fast-cn
lab01(config)#crypto key generate rsa
The name for the keys will be: lab01.fast-cn
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

lab01(config)#ip s
*Mar 1 0:25:35.449: %SSH-5-ENABLED: SSH 1.99 has been enabled
% Ambiguous command: "ip s"
lab01(config)#ip ssh version 2
lab01(config)#line vty 0 15
lab01(config-line)#transport input ssh
lab01(config-line)#
```

Protocol working on it. By default, username is admin. We can apply any sort of configuration on our switch from our PC.



Now, if we want to change the username from admin to something else, we will do it as follows:

```
lab01(config)#username CS-FAST secret abc
lab01(config)#line vty 0 15
lab01(config-line)#login local
lab01(config-line)#
```

Now we will do the following on our PCs:

```
C:\>ssh -l CS-FAST 192.168.1.1

Password:


lab01>
```